

BUSINESS EXCELLENCE ANALYTICS REPORT

(Furniture, Office Supplies & Technology Performance Analysis)

1. Project Overview and Objective

The dataset contains common data quality issues such as missing values, duplicate records, incorrect data entries, and inconsistent formatting. Using Excel, Power Query, and Power BI, the data is cleaned, transformed, and analyzed. Since the dataset consists of multiple related tables, data modeling is performed by creating relationships between the tables. DAX measures, KPI Cards, Gauge Charts, Matrix, and other visualizations are used to build an interactive dashboard for sales analysis and business decision-making.

2. Data Sources

- **Source Description and Timeline:**Google Dataset Search and (2023 - 2025)
- **Domain:** Retail

3. Problem Statement

This project analyzing Retail Sales data from 2023 to 2025 to identify sales trends, customer purchasing patterns, product performance, and profit insights

4. Attribute (Column /Features) Details:

Sales Sheet 1:

Attribute Name	Data Type	Description
Order ID	Integer / String	Unique identifier for each orders
Order Date	Date Data Type	Order Date for each orders
Ship Date	Date Data Type	Shipped date for each orders
Customer ID	Integer / String	Unique identifier for each Customers
Product ID	Integer / String	Unique identifier for each product
Sales	Decimal Number	Sales amount for each sales
Quantity	Whole Number	Quantity for each Orders
Discount	Decimal Number	Discount for each Orders
Ship Mode	Text/Categorical	Shipping mode used for deliver the order
Sales Channel	Text	Sales channel used for order
Returned	Categorical	Shows whether the sale was returned

Customer Sheet 2:

Attribute Name	Data Type	Description
Customer ID	Integer / String	Unique identifier for each Customers
Segment	Categorical	Segment for each customer type
State	Categorical	State of the each customer
Customer Rating	Categorical	Rating given by the customer

Product Sheet 3:

Attribute Name	Data Type	Description
Product ID	Integer / String	Unique identifier for each product
Category	Categorical	Category for each product
Sub-Category	Categorical	Sub Category for each product
Product Name	Text	Name of each Product

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

1.Data Cleaning & Transformation: Raw data changed to standardized and structured table format

	A	B	C	D	E	F	G	H	I	J	K
	Order ID	Order Date	Ship Date	Customer ID	Product ID	Sales	Quantity	Discount	Ship Mode	Sales Channel	Returned
1	ORD00001	01/26/2023	2023-01-29	CUST0211	PROD002	₹ 4,075	6	0.2		mobile App	yes
2	ORD00002	10/25/2023	2023-10-29	CUST0178	PROD049	₹ 4,035	5	0	First Class	mobile App	no
3	ORD00003	25-11-2024	2024-11-30	CUST0069	PROD053	₹ 1,676	10	0.1		Mobile App	yes
4	ORD00004	12/01/2024	2024-12-04	CUST0053	PROD080	₹ 4,538	3	0.3	First Class	Online	yes
5	ORD00005	11/16/2023	2023-11-18	CUST0157	PROD004	₹ 783	7	0.1	second Class	store	no
6	ORD00006	25-11-2024	2024-11-30	CUST0118	PROD012	₹ 1,026	9	0.1	Standard Class	Store	no
7	ORD00007	01/07/2025	2025-01-10	CUST0202	PROD007	₹ 352	2	0	second Class	Store	yes
8	ORD00008	09/20/2024	2024-09-21	CUST0186	PROD036	₹ 2,064	3	0.3	second Class	Store	no
9	ORD00009	02-08-2023	2023-08-09	CUST0020	PROD022	₹ 3,637	5	0.2	Standard Class	Store	no
10	ORD00010	07/16/2023	2023-07-17	CUST0177	PROD028	₹ 569	8	0.1	second Class	online	no
11	ORD00011	01/11/2023	2023-01-16	CUST0004	PROD068	₹ 4,248	4	0.3		Mobile App	no
12	ORD00012	26-03-2024	2024-03-30	CUST0130	PROD077	₹ 1,533	1	0.1	First Class	Store	no
13	ORD00013	03/06/2025	2025-03-12	CUST0027	PROD085	₹ 1,616	5	0.1	second Class	Store	yes
14	ORD00014	02/27/2024	2024-02-29	CUST0028	PROD035	₹ 2,865	3	0.3	First Class	Online	yes
15	ORD00015	05-05-2024	2024-05-08	CUST0223	PROD038	₹ 3,781	10	0.1	second Class	online	no
16	ORD00016	09/15/2023	2023-09-21	CUST0223	PROD009	₹ 827	6	0.2	second Class	Mobile App	no
17	ORD00017	10/05/2023	2023-10-06	CUST0239	PROD066	₹ 4,480	10	0	second Class	online	no
18	ORD00018	15-04-2025	2025-04-20	CUST0006	PROD042	₹ 543	3	0.1	First Class	Mobile App	no
19	ORD00019	04/23/2023	2023-04-30	CUST0135	PROD075	₹ 2,608	2	0.1	second Class	Mobile App	yes

	A	B	C	D
	Customer ID	Segment	State	Customer Rating
1	CUST0001	Home Office	MAHARASHTRA	4
2	CUST0002	consumer	telangana	1
3	CUST0003	Consumer	tElanganA	5
4	CUST0004	Consumer	Kerala	3
5	CUST0005	home Office	maharashtra	4
6	CUST0006	Corporate	Tamil Nadu	4
7	CUST0007	Consumer	TAMIL NADU	3
8	CUST0008	Home Office	karnataka	2
9	CUST0009	Home Office	Tamil Nadu	5
10	CUST0010	Corporate	Tamil Nadu	5
11	CUST0011	consumer	Karnataka	4
12	CUST0012	Home Office	Kerala	1
13	CUST0013	Home Office	Maharashtra	2
14	CUST0014	corporate	Karnataka	1
15	CUST0015	Corporate	Maharashtra	2
16	CUST0016	Consumer	Karnataka	3
17	CUST0017	corporate	Telangana	5
18	CUST0018	Consumer	Maharashtra	3
19	CUST0019	Home Office	Tamil Nadu	3

Product ID	Category	Sub-Category	Product Name
PROD001	Office Supplies	Accessories	Product 1
PROD002	Office Supplies	Binders	Product 2
PROD003	Furniture	Chairs	Product 3
PROD004	Technology	Binders	Product 4
PROD005	Technology	Storage	Product 5
PROD006	Furniture	Binders	Product 6
PROD007	Furniture	Storage	Product 7
PROD008	Furniture	Phones	Product 8
PROD009	Technology	Storage	Product 9
PROD010	Furniture	Tables	Product 10
PROD011	Office Supplies	Chairs	Product 11
PROD012	Office Supplies	Chairs	Product 12
PROD013	Furniture	Accessories	Product 13
PROD014	Technology	Chairs	Product 14
PROD015	Furniture	Tables	Product 15
PROD016	Office Supplies	Storage	Product 16
PROD017	Furniture	Binders	Product 17
PROD018	Office Supplies	Accessories	Product 18
PROD019	Office Supplies	Accessories	Product 19

< > Sales Customers **Products** +

2.Data Cleaning:

From sales table I have used **LEFT,RIGHT,MID** functions on the **Order ID** Column to Extract the Id text into 3 separate columns

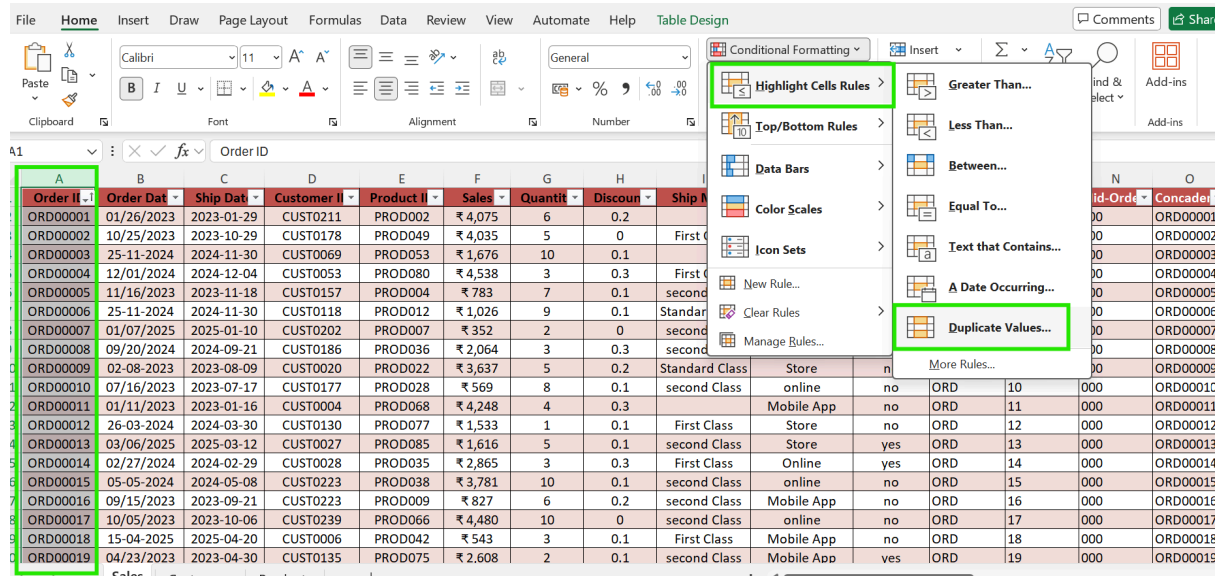
Next I have applied **CONCATENATE** Functions on the those extracted separated columns merged to one single column

Then used the **PROPER** functions on the **RETURNED, SALES CHANNEL,SHIP MODE** Columns to correct the text formatting and capitalize the first letter of each word

Left - Ord	Right - Ord	Mid-Ord	Concaten	Proper -Returned	Proper - Sales Channel	Proper-Ship mode
ORD	01	000	ORD00001	Yes	Mobile App	
ORD	02	000	ORD00002	No	Mobile App	First Class
ORD	03	000	ORD00003	Yes	Mobile App	
ORD	04	000	ORD00004	Yes	Online	First Class
ORD	05	000	ORD00005	No	Store	Second Class
ORD	06	000	ORD00006	No	Store	Standard Class
ORD	07	000	ORD00007	Yes	Store	Second Class
ORD	08	000	ORD00008	No	Store	Second Class
ORD	09	000	ORD00009	No	Store	Standard Class
ORD	10	000	ORD00010	No	Online	Second Class
ORD	11	000	ORD00011	No	Mobile App	
ORD	12	000	ORD00012	No	Store	First Class
ORD	13	000	ORD00013	Yes	Store	Second Class
ORD	14	000	ORD00014	Yes	Online	First Class
ORD	15	000	ORD00015	No	Online	Second Class
ORD	16	000	ORD00016	No	Mobile App	Second Class
ORD	17	000	ORD00017	No	Online	Second Class
ORD	18	000	ORD00018	No	Mobile App	First Class
ORD	19	000	ORD00019	Yes	Mobile App	Second Class

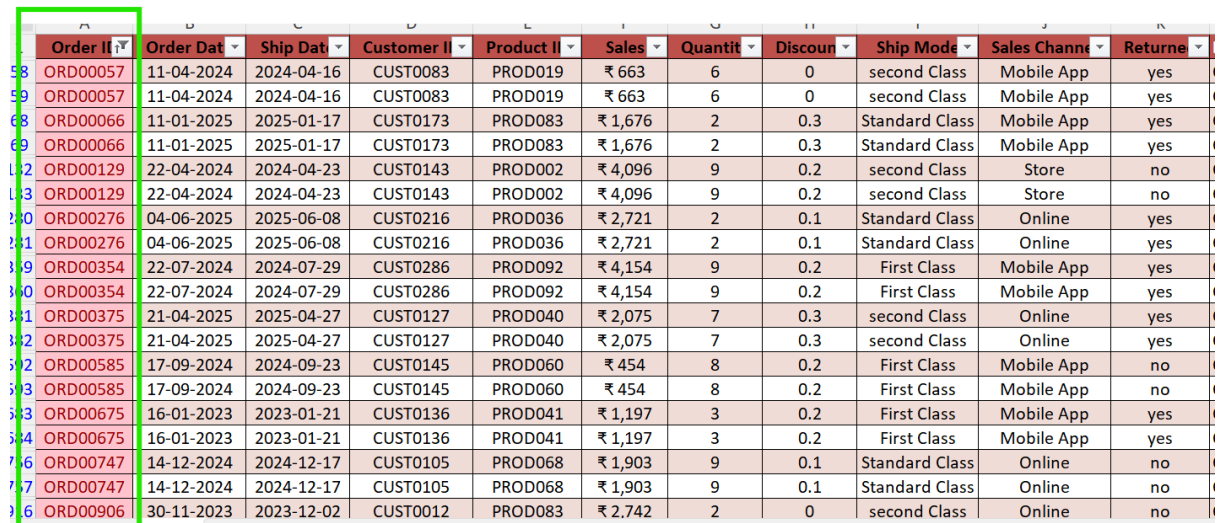
Find Duplicate values

In the Sales table, I used **Conditional Formatting** and applied the **Duplicate Values** option to the Order ID column to identify duplicate orders

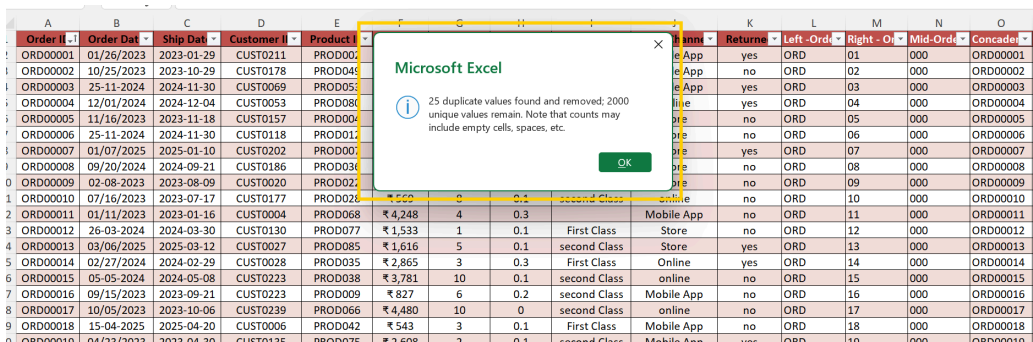
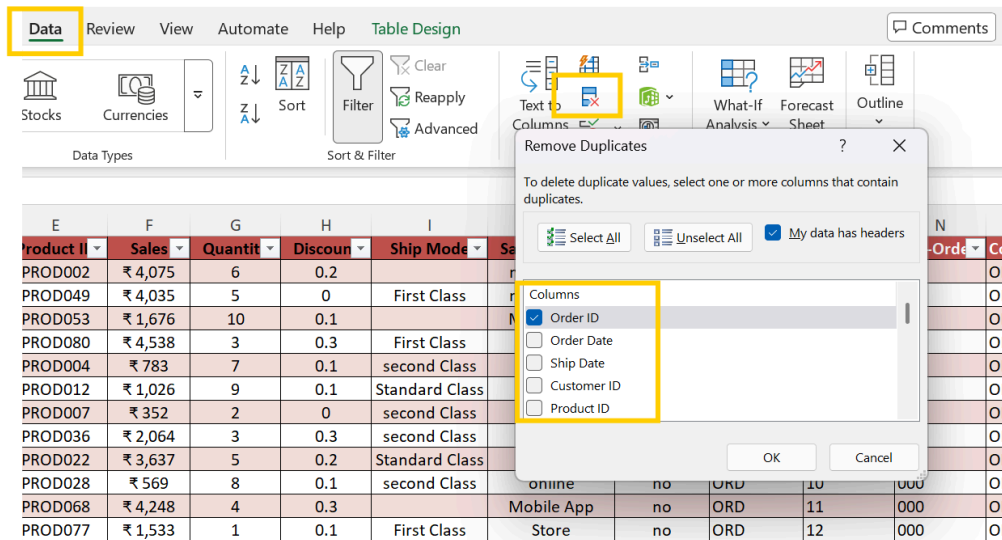


Order ID	Order Date	Ship Date	Customer ID	Product ID	Sales	Quantity	Discount	Ship Mode	Sales Channel	Return Status
ORD00001	01/26/2023	2023-01-29	CUST0211	PROD002	₹ 4,075	6	0.2	Standard Class	Store	no
ORD00002	10/25/2023	2023-10-29	CUST0178	PROD049	₹ 4,035	5	0	First Class	online	no
ORD00003	25-11-2024	2024-11-30	CUST0069	PROD053	₹ 1,676	10	0.1	Standard Class	Mobile App	yes
ORD00004	12/01/2024	2024-12-04	CUST0053	PROD080	₹ 4,538	3	0.3	Standard Class	Mobile App	yes
ORD00005	11/16/2023	2023-11-18	CUST0157	PROD004	₹ 783	7	0.1	Standard Class	Mobile App	yes
ORD00006	25-11-2024	2024-11-30	CUST0118	PROD012	₹ 1,026	9	0.1	Standard Class	Mobile App	yes
ORD00007	01/07/2025	2025-01-10	CUST0202	PROD007	₹ 352	2	0	Standard Class	Mobile App	yes
ORD00008	09/20/2024	2024-09-21	CUST0186	PROD036	₹ 2,064	3	0.3	Standard Class	Mobile App	yes
ORD00009	02-08-2023	2023-08-09	CUST0020	PROD022	₹ 3,637	5	0.2	Standard Class	Store	no
ORD00010	07/16/2023	2023-07-17	CUST0177	PROD028	₹ 569	8	0.1	Standard Class	online	no
ORD00011	01/11/2023	2023-01-16	CUST0004	PROD068	₹ 4,248	4	0.3	Standard Class	Mobile App	no
ORD00012	26-03-2024	2024-03-30	CUST0130	PROD077	₹ 1,533	1	0.1	First Class	Store	no
ORD00013	03/06/2025	2025-03-12	CUST0027	PROD085	₹ 1,616	5	0.1	Standard Class	Store	yes
ORD00014	02/27/2024	2024-02-29	CUST0028	PROD035	₹ 2,865	3	0.3	First Class	Online	yes
ORD00015	05-05-2024	2024-05-08	CUST0223	PROD038	₹ 3,781	10	0.1	Standard Class	online	no
ORD00016	09/15/2023	2023-09-21	CUST0223	PROD009	₹ 827	6	0.2	Standard Class	Mobile App	no
ORD00017	10/05/2023	2023-10-06	CUST0239	PROD066	₹ 4,480	10	0	Standard Class	online	no
ORD00018	15-04-2025	2025-04-20	CUST0006	PROD042	₹ 543	3	0.1	First Class	Mobile App	no
ORD00019	04/23/2023	2023-04-30	CUST0135	PROD075	₹ 2,608	2	0.1	Standard Class	Mobile App	yes

Then Filtered the duplicate **Order IDs** using the **Filter by Color** option



Order ID	Order Date	Ship Date	Customer ID	Product ID	Sales	Quantity	Discount	Ship Mode	Sales Channel	Return Status
ORD00001	01-04-2024	2024-04-16	CUST0083	PROD019	₹ 663	6	0	second Class	Mobile App	yes
ORD00007	11-04-2024	2024-04-16	CUST0083	PROD019	₹ 663	6	0	second Class	Mobile App	yes
ORD00066	11-01-2025	2025-01-17	CUST0173	PROD083	₹ 1,676	2	0.3	Standard Class	Mobile App	yes
ORD00066	11-01-2025	2025-01-17	CUST0173	PROD083	₹ 1,676	2	0.3	Standard Class	Mobile App	yes
ORD00129	22-04-2024	2024-04-23	CUST0143	PROD002	₹ 4,096	9	0.2	second Class	Store	no
ORD00129	22-04-2024	2024-04-23	CUST0143	PROD002	₹ 4,096	9	0.2	second Class	Store	no
ORD00276	04-06-2025	2025-06-08	CUST0216	PROD036	₹ 2,721	2	0.1	Standard Class	Online	yes
ORD00276	04-06-2025	2025-06-08	CUST0216	PROD036	₹ 2,721	2	0.1	Standard Class	Online	yes
ORD00354	22-07-2024	2024-07-29	CUST0286	PROD092	₹ 4,154	9	0.2	First Class	Mobile App	yes
ORD00354	22-07-2024	2024-07-29	CUST0286	PROD092	₹ 4,154	9	0.2	First Class	Mobile App	yes
ORD00375	21-04-2025	2025-04-27	CUST0127	PROD040	₹ 2,075	7	0.3	second Class	Online	yes
ORD00375	21-04-2025	2025-04-27	CUST0127	PROD040	₹ 2,075	7	0.3	second Class	Online	yes
ORD00585	17-09-2024	2024-09-23	CUST0145	PROD060	₹ 454	8	0.2	First Class	Mobile App	no
ORD00585	17-09-2024	2024-09-23	CUST0145	PROD060	₹ 454	8	0.2	First Class	Mobile App	no
ORD00675	16-01-2023	2023-01-21	CUST0136	PROD041	₹ 1,197	3	0.2	First Class	Mobile App	yes
ORD00675	16-01-2023	2023-01-21	CUST0136	PROD041	₹ 1,197	3	0.2	First Class	Mobile App	yes
ORD00747	14-12-2024	2024-12-17	CUST0105	PROD068	₹ 1,903	9	0.1	Standard Class	Online	no
ORD00747	14-12-2024	2024-12-17	CUST0105	PROD068	₹ 1,903	9	0.1	Standard Class	Online	no
ORD00906	30-11-2023	2023-12-02	CUST0012	PROD083	₹ 2,742	2	0	second Class	Online	no



In the **Data ribbon**, I applied the **Duplicate Values** option to the Order ID column. I identified **25 duplicate values** and removed them.

3.Formatting Date Properly

In the Sales table, the **Order Date** and **Ship Date** were not formatted properly. I applied the Delimiter option and used the **DATE, IF**, and **VALUE** functions to convert the dates into the proper **DD-MM-YYYY** format. Now, both the Order Date and Ship Date are cleaned properly

Order Date formatted	Ship date formatted	Proper Order Date	Proper ship date
01-26-2023	2023-01-29	26-01-2023	29-01-2023
10-25-2023	2023-10-29	25-10-2023	29-10-2023
25-11-2024	2024-11-30	25-11-2024	30-11-2024
12-01-2024	2024-12-04	01-12-2024	04-12-2024
11-16-2023	2023-11-18	16-11-2023	18-11-2023
25-11-2024	2024-11-30	25-11-2024	30-11-2024
01-07-2025	2025-01-10	07-01-2025	10-01-2025
09-20-2024	2024-09-21	20-09-2024	21-09-2024
02-08-2023	2023-08-09	08-02-2023	09-08-2023
07-16-2023	2023-07-17	16-07-2023	17-07-2023
01-11-2023	2023-01-16	11-01-2023	16-01-2023
26-03-2024	2024-03-30	26-03-2024	30-03-2024
03-06-2025	2025-03-12	06-03-2025	12-03-2025
02-27-2024	2024-02-29	27-02-2024	29-02-2024
05-05-2024	2024-05-08	05-05-2024	08-05-2024
09-15-2023	2023-09-21	15-09-2023	21-09-2023
10-05-2023	2023-10-06	05-10-2023	06-10-2023
15-04-2025	2025-04-20	15-04-2025	20-04-2025

Proper - Sales Channel	Proper-Ship mode	Order Date formatted	Ship date formatted	Proper Order Date	Proper ship date	Column13	Column14	Column15	Column16
Mobile App		01-26-2023	2023-01-29	26-01-2023	29-01-2023				
Mobile App	First Class	10-25-2023	2023-10-29	25-10-2023	29-10-2023				
Mobile App		25-11-2024	2024-11-30	25-11-2024	30-11-2024				
Online	First Class	12-01-2024	2024-12-04	01-12-2024	04-12-2024				
Store	Second Class	11-16-2023	2023-11-18	16-11-2023	18-11-2023				
Store	Standard Class	25-11-2024	2024-11-30	25-11-2024	30-11-2024				
Store	Second Class	01-07-2025	2025-01-10	07-01-2025	10-01-2025				
Store	Second Class	09-20-2024	2024-09-21	20-09-2024	21-09-2024				
Store	Standard Class	02-08-2023	2023-08-09	08-02-2023	09-08-2023				
Online	Second Class	07-16-2023	2023-07-17	16-07-2023	17-07-2023				
Mobile App		01-11-2023	2023-01-16	11-01-2023	16-01-2023				
Store	First Class	26-03-2024	2024-03-30	26-03-2024	30-03-2024				
Store	Second Class	03-06-2025	2025-03-12	06-03-2025	12-03-2025				
Online	First Class	02-27-2024	2024-02-29	27-02-2024	29-02-2024				
Online	Second Class	05-05-2024	2024-05-08	05-05-2024	08-05-2024				
Mobile App	Second Class	09-15-2023	2023-09-21	15-09-2023	21-09-2023				

4.Added Columns

Then, in the Customer table, I applied Nested IF conditions to the Customer Rating column. I categorized the ratings as follows: 5 – Excellent, 4 – Good, 3 – Average, 2 – Poor, and 1 – Very Poor

Customer ID	Segment	State	Customer Rating	Rating status	Proper Segement	Proper State	Column4	Column5	Column6	Column7	Column8	Column9	Column10
CUST0001	Home Office	MAHARASHTRA	4	Good	Home Office	Maharashtra							
CUST0002	consumer	telangana	1	Very Poor	Consumer	Telangana							
CUST0003	Consumer	tElanganA	5	Excellent	Consumer	Telangana							
CUST0004	Consumer	Kerala	3	Average	Consumer	Kerala							
CUST0005	home Office	maharashtra	4	Good	Home Office	Maharashtra							
CUST0006	Corporate	Tamil Nadu	4	Good	Corporate	Tamil Nadu							
CUST0007	Consumer	TAMIL NADU	3	Average	Consumer	Tamil Nadu							
CUST0008	Home Office	karnataka	2	Poor	Home Office	Karnataka							
CUST0009	Home Office	Tamil Nadu	5	Excellent	Home Office	Tamil Nadu							
CUST0010	Corporate	Tamil Nadu	5	Excellent	Corporate	Tamil Nadu							
CUST0011	consumer	Karnataka	4	Good	Consumer	Karnataka							
CUST0012	Home Office	Kerala	1	Very Poor	Home Office	Kerala							
CUST0013	Home Office	Maharashtra	2	Poor	Home Office	Maharashtra							
CUST0014	corporate	Karnataka	1	Very Poor	Corporate	Karnataka							
CUST0015	Corporate	Maharashtra	2	Poor	Corporate	Maharashtra							
CUST0016	Consumer	Karnataka	3	Average	Consumer	Karnataka							
CUST0017	corporate	Telangana	5	Excellent	Corporate	Telangana							

Then, the text in the **Segment and State columns** was not formatted properly, so I used the **PROPER** function to correct the text formatting.

5.Extra Spaces and Inconsistencies

In the Products table, the **Category and Sub-Category columns** text had extra spaces and inconsistencies. So, I used the **TRIM** function to remove the extra spaces and correct the text.

2

Product ID	Category	Sub-Category	Product Name	Trimmed Category	Trimmed Sub-Category	Color
PROD001	Office Supplies	Accessories	Product 1	Office Supplies	Accessories	
PROD002	Office Supplies	Binders	Product 2	Office Supplies	Binders	
PROD003	Furniture	Chairs	Product 3	Furniture	Chairs	
PROD004	Technology	Binders	Product 4	Technology	Binders	
PROD005	Technology	Storage	Product 5	Technology	Storage	
PROD006	Furniture	Binders	Product 6	Furniture	Binders	
PROD007	Furniture	Storage	Product 7	Furniture	Storage	
PROD008	Furniture	Phones	Product 8	Furniture	Phones	
PROD009	Technology	Storage	Product 9	Technology	Storage	
PROD010	Furniture	Tables	Product 10	Furniture	Tables	
PROD011	Office Supplies	Chairs	Product 11	Office Supplies	Chairs	
PROD012	Office Supplies	Chairs	Product 12	Office Supplies	Chairs	
PROD013	Furniture	Accessories	Product 13	Furniture	Accessories	
PROD014	Technology	Chairs	Product 14	Technology	Chairs	
PROD015	Furniture	Tables	Product 15	Furniture	Tables	
PROD016	Office Supplies	Storage	Product 16	Office Supplies	Storage	
PROD017	Furniture	Binders	Product 17	Furniture	Binders	
PROD018	Office Supplies	Accessories	Product 18	Office Supplies	Accessories	
PROD019	Office Supplies	Accessories	Product 19	Office Supplies	Accessories	

Sales Customers **Products**

6.Power Query

Created a product-wise median Sales table.Merged it with the Sales table using Product ID.Expanded the median Sales column.Used a Custom Column to fill the missing Sales values with the median.

Product ID	1.2 median products
PROD002	2064.83
PROD049	3321.785
PROD053	1676.01
PROD080	3422.905
PROD004	2911.4
PROD012	2031.725
PROD007	2514.42
PROD036	2676.745
PROD022	3070.02
PROD028	3016.94
PROD068	2761.11
PROD077	1884.41
PROD085	2211.43
PROD035	2120.04
PROD038	2760.43
PROD009	3199.095
PROD066	3189.9
PROD042	2383.33
PROD075	2713.41
PROD021	2379.64
PROD047	1421.12

PROPERTIES

Name
Table1 (2)

All Properties

APPLIED STEPS

Source
Changed Type
X median products

Close Query Manage Columns Reduce Rows Sort Transform Combine Parameters Data Sources New Query

Queries [3]
Table1
Table1 (2)
Table1 (3)

fx = Table.ExpandTableColumn(#"Merged Queries", "Table1 (2)", ("Product ID", "median products"),

	matted	Proper Order Date	Proper ship date	Table1 (2).Product ID	Table1 (2).median products
1	29-01-2023	26-01-2023 00:00:00	29-01-2023	PROD002	2064.83
2	11-05-2023	04-05-2023 00:00:00	11-05-2023	PROD002	2064.83
3	05-11-2023	11-04-2023 00:00:00	05-11-2023	PROD002	2064.83
4	29-10-2023	25-10-2023 00:00:00	29-10-2023	PROD049	3321.785
5	11-08-2024	04-08-2024 00:00:00	11-08-2024	PROD049	3321.785
6	30-11-2024	25-11-2024 00:00:00	30-11-2024	PROD053	1676.01
7	04-12-2024	01-12-2024 00:00:00	04-12-2024	PROD080	3422.905
8	01-09-2023	28-08-2023 00:00:00	01-09-2023	PROD080	3422.905
9	15-09-2023	09-12-2023 00:00:00	15-09-2023	PROD080	3422.905
10	18-11-2023	16-11-2023 00:00:00	18-11-2023	PROD004	2911.4
11	30-11-2024	25-11-2024 00:00:00	30-11-2024	PROD012	2031.725
12	19-11-2023	11-12-2023 00:00:00	19-11-2023	PROD012	2031.725
13	26-02-2025	19-02-2025 00:00:00	26-02-2025	PROD012	2031.725
14	03-11-2023	27-10-2023 00:00:00	03-11-2023	PROD012	2031.725
15	25-10-2024	20-10-2024 00:00:00	25-10-2024	PROD012	2031.725
16	10-01-2025	07-01-2025 00:00:00	10-01-2025	PROD007	2514.42
17	21-09-2024	20-09-2024 00:00:00	21-09-2024	PROD036	2676.745
18	09-08-2023	08-02-2023 00:00:00	09-08-2023	PROD022	3070.02
19	17-07-2023	16-07-2023 00:00:00	17-07-2023	PROD028	3016.94
20	16-01-2023	11-01-2023 00:00:00	16-01-2023	PROD068	2761.11
21					

Query Settings

PROPERTIES
Name
Table1

APPLIED STEPS
Source
Changed Type
Merged Queries
Expanded Table1 (2)
Added Custom
final sales
Changed Type1
Replaced Value
Capitalized Each Word

File Home Transform Add Column View

Close & Load Refresh Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Whole Number Merge Queries Append Queries Combine Files Manage Parameters Data source settings New Source Recent Sources Enter Data

Queries [3]
Table1
Table1 (2)
Table1 (3)

fx = Table.ReorderColumns(#"Added Custom",("Order ID", "Order Date", "Ship Date", "Customer ID",

	Customer ID	Product ID	final sales	Sales	Quantity
1	IT0211	PROD002	4075.4	4075.4	6
2	IT0173	PROD002	327.45	327.45	8
3	IT0275	PROD002	1262.75	1262.75	3
4	IT0178	PROD049	4035.22	4035.22	5
5	IT0095	PROD049	3968.01	3968.01	8
6	IT0069	PROD053	1676.01	1676.01	10
7	IT0053	PROD080	4538.4	4538.4	3
8	IT0290	PROD080	1672.66	1672.66	7
9	IT0089	PROD080	3964.58	3964.58	4
10	IT0157	PROD004	783.4	783.4	7
11	IT0118	PROD012	1026.47	1026.47	9
12	IT0046	PROD012	1835.05	1835.05	9
13	IT0008	PROD012	1665.1	1665.1	5
14	IT0060	PROD012	1427.32	1427.32	6
15	IT0020	PROD012	885.43	885.43	4

Query Settings

PROPERTIES
Name
Table1

APPLIED STEPS
Source
Changed Type
Merged Queries
Expanded Table1 (2)
Added Custom
final sales
Changed Type1
Replaced Value
Capitalized Each Word

In the Ship Mode column, I identified the null values and replaced them using the mode value, which was First Class

Close Query Manage Columns Reduce Rows Sort Transform Combine Parameters Data Sources New Query

Queries [3]
Table1
Table1 (2)
Table1 (3)

fx = Table.TransformColumns(#"Replaced Value",({"Ship Mode", Text.Proper, type text}))

	Sales	Quantity	Discount	Ship Mode	Sales Channel
1	4075.4	6	0	First Class	mobile App
2	327.45	8	0	Second Class	Online
3	1262.75	3	0	First Class	Online
4	4035.22	5	0	First Class	mobile App
5	3968.01	8	0	First Class	Mobile App
6	1676.01	10	0	First Class	Mobile App
7	4538.4	3	0	First Class	Online
8	1672.66	7	0	First Class	Mobile App
9	3964.58	4	0	Standard Class	Mobile App
10	783.4	7	0	Second Class	store
11	1026.47	9	0	Standard Class	Store
12	1835.05	9	0	First Class	Mobile App
13	1665.1	5	0	Second Class	Online
14	1427.32	6	0	Standard Class	Online
15	885.43	4	0	Standard Class	Online
16	352.44	2	0	Second Class	Store
17	2063.72	3	0	Second Class	Store
18	3637.24	5	0	Standard Class	Store
19	569	8	0	Second Class	online
20	4248.01	4	0	First Class	Mobile App

Query Settings

PROPERTIES
Name
Table1

APPLIED STEPS
Source
Changed Type
Merged Queries
Expanded Table1 (2)
Added Custom
final sales
Changed Type1
Replaced Value
Capitalized Each Word

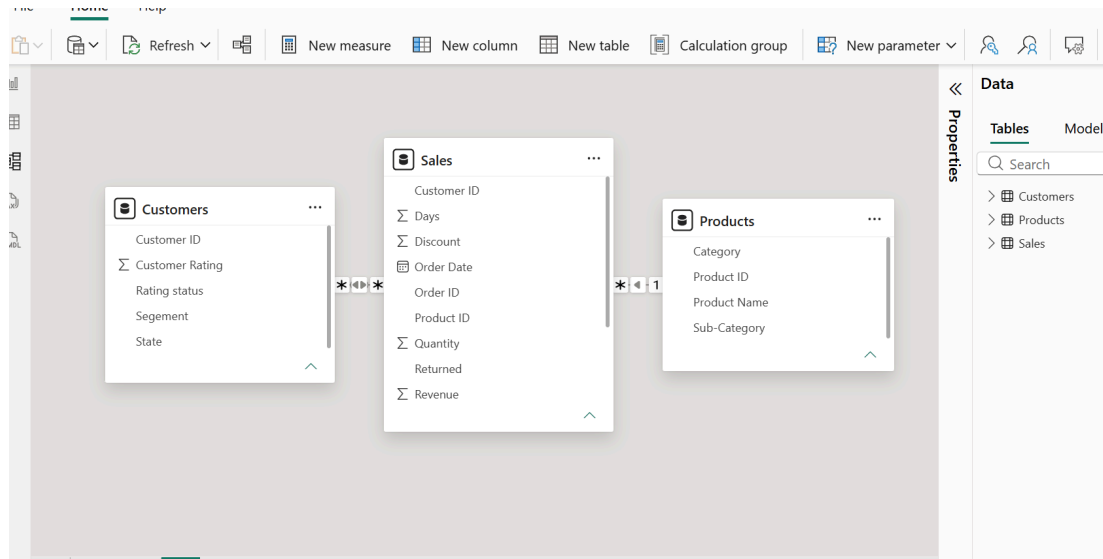
7.I calculated the Revenue using the Sales Amount and Discount columns.

Sales	Quantity	Discount	Ship Mode	Returned	Sales Channel	Revenue	Order Date	Ship Date	Days
₹ 4,075	6	0.2	First Class	Yes	Mobile App	₹ 3,260	26-01-2023	29-01-2023	3
₹ 4,035	5	0	First Class	No	Mobile App	₹ 4,035	25-10-2023	29-10-2023	4
₹ 1,676	10	0.1	First Class	Yes	Mobile App	₹ 1,508	25-11-2024	30-11-2024	5
₹ 4,538	3	0.3	First Class	Yes	Online	₹ 3,177	01-12-2024	04-12-2024	3
₹ 783	7	0.1	Second Class	No	Store	₹ 705	16-11-2023	18-11-2023	2
₹ 1,026	9	0.1	Standard Class	No	Store	₹ 924	25-11-2024	30-11-2024	5
₹ 352	2	0	Second Class	Yes	Store	₹ 352	07-01-2025	10-01-2025	3
₹ 2,064	3	0.3	Second Class	No	Store	₹ 1,445	20-09-2024	21-09-2024	1
₹ 3,637	5	0.2	Standard Class	No	Store	₹ 2,910	08-02-2023	09-08-2023	7
₹ 569	8	0.1	Second Class	No	Online	₹ 512	16-07-2023	17-07-2023	1
₹ 4,248	4	0.3	First Class	No	Mobile App	₹ 2,974	11-01-2023	16-01-2023	5
₹ 1,533	1	0.1	First Class	No	Store	₹ 1,380	26-03-2024	30-03-2024	4
₹ 1,616	5	0.1	Second Class	Yes	Store	₹ 1,455	06-03-2025	12-03-2025	6
₹ 2,865	3	0.3	First Class	Yes	Online	₹ 2,005	27-02-2024	29-02-2024	2
₹ 3,781	10	0.1	Second Class	No	Online	₹ 3,403	05-05-2024	08-05-2024	3
₹ 827	6	0.2	Second Class	No	Mobile App	₹ 662	15-09-2023	21-09-2023	6
₹ 4,480	10	0	Second Class	No	Online	₹ 4,480	05-10-2023	06-10-2023	1
₹ 543	3	0.1	First Class	No	Mobile App	₹ 489	15-04-2025	20-04-2025	5
₹ 2,608	2	0.1	Second Class	Yes	Mobile App	₹ 2,347	23-04-2023	30-04-2023	7

Finally I have calculated the number of days between the Order Date and Ship Date to determine the shipping duration

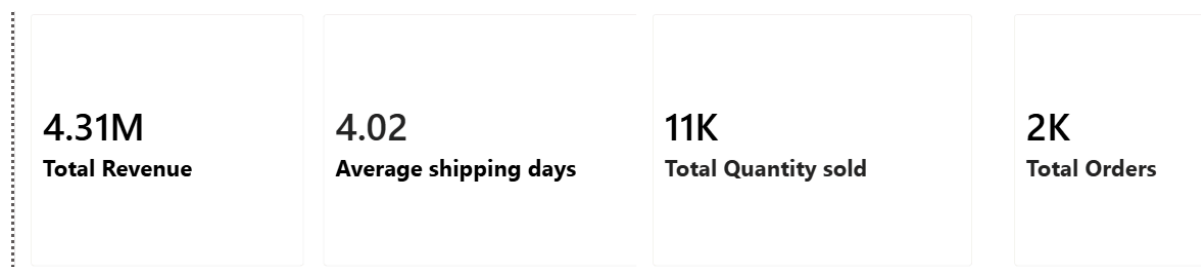
Quantity	Discount	Ship Mode	Returned	Sales Channel	Revenue	Order Date	Ship Date	Days
6	0.2	First Class	Yes	Mobile App	₹ 3,260	26-01-2023	29-01-2023	3
5	0	First Class	No	Mobile App	₹ 4,035	25-10-2023	29-10-2023	4
10	0.1	First Class	Yes	Mobile App	₹ 1,508	25-11-2024	30-11-2024	5
3	0.3	First Class	Yes	Online	₹ 3,177	01-12-2024	04-12-2024	3
7	0.1	Second Class	No	Store	₹ 705	16-11-2023	18-11-2023	2
9	0.1	Standard Class	No	Store	₹ 924	25-11-2024	30-11-2024	5
2	0	Second Class	Yes	Store	₹ 352	07-01-2025	10-01-2025	3
3	0.3	Second Class	No	Store	₹ 1,445	20-09-2024	21-09-2024	1
5	0.2	Standard Class	No	Store	₹ 2,910	02-08-2023	09-08-2023	7
8	0.1	Second Class	No	Online	₹ 512	16-07-2023	17-07-2023	1
4	0.3	First Class	No	Mobile App	₹ 2,974	11-01-2023	16-01-2023	5
1	0.1	First Class	No	Store	₹ 1,380	26-03-2024	30-03-2024	4
5	0.1	Second Class	Yes	Store	₹ 1,455	06-03-2025	12-03-2025	6
3	0.3	First Class	Yes	Online	₹ 2,005	27-02-2024	29-02-2024	2
10	0.1	Second Class	No	Online	₹ 3,403	05-05-2024	08-05-2024	3
6	0.2	Second Class	No	Mobile App	₹ 662	15-09-2023	21-09-2023	6
10	0	Second Class	No	Online	₹ 4,480	05-10-2023	06-10-2023	1
3	0.1	First Class	No	Mobile App	₹ 489	15-04-2025	20-04-2025	5
2	0.1	Second Class	Yes	Mobile App	₹ 2,347	23-04-2023	30-04-2023	7

8 Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.



First, I created data relationships between the Customer, Sales, and Product tables in Power BI and checked that the relationships were active

I have created new Cards using measures to display Total Revenue, Average Shipping Days, Total orders and Total Quantity Sold



Created a pie chart to show the revenue distribution across different states

Created a clustered column chart to analyze revenue across different categories and sub-categories

Created a donut chart to show the revenue distribution across different customer segments.

Added a tooltip to display Revenue and Discount details when hovering over the chart

Created a line chart to compare monthly revenue for returned and non-returned orders

Created a clustered bar chart to compare revenue across different ship modes based on return status

Created a stacked column chart to compare quantity sold across categories and sub-categories

Created a scatter chart to compare revenue and quantity by rating status.

Created a KPI card to compare total revenue with an assumed sales target and track the revenue trend over time

Created a gauge chart to compare total revenue with the assumed sales target

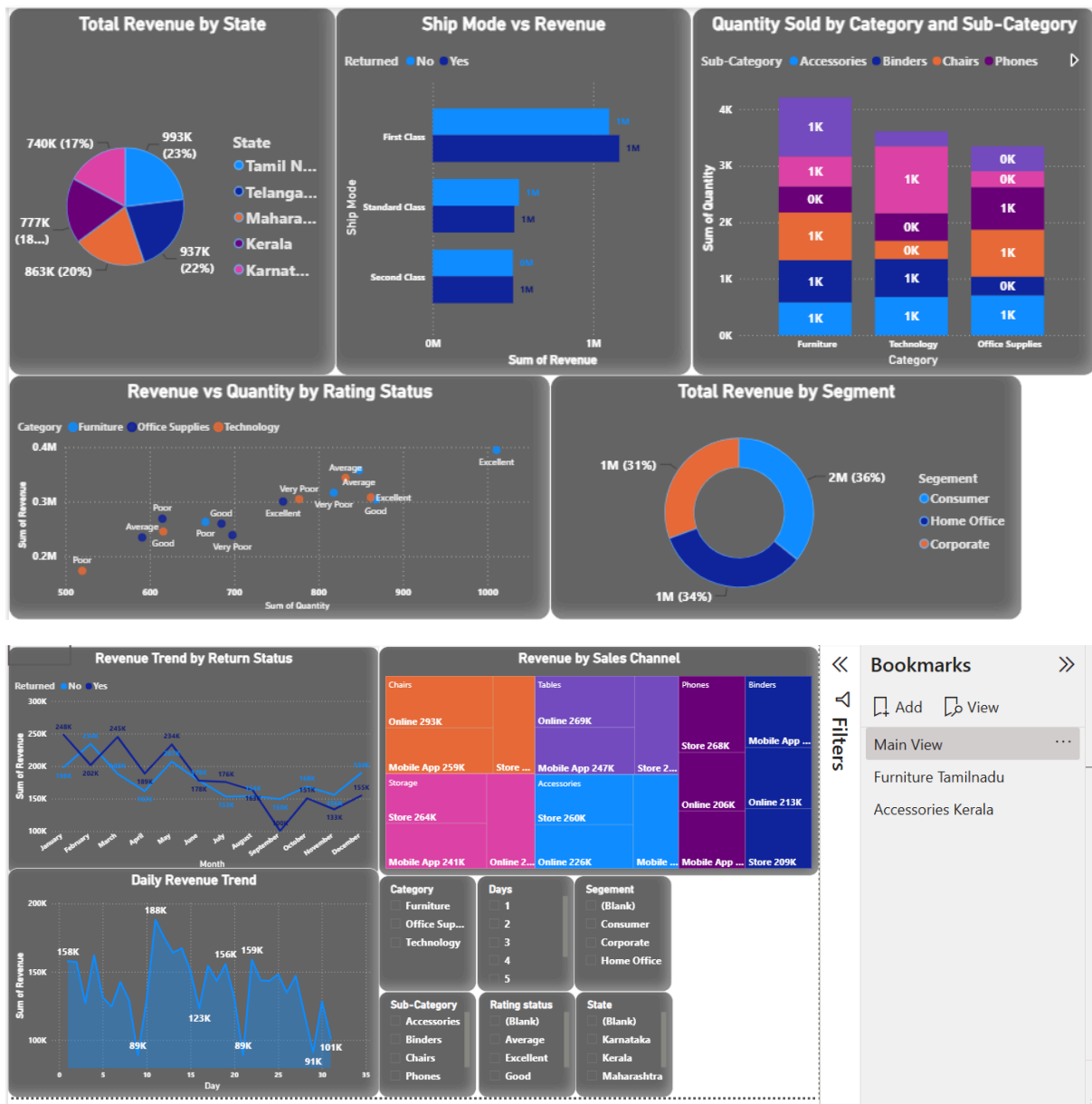
The treemap shows revenue by sales channel and sub-category

The area chart shows daily revenue and helps identify the sales trend over time.

Created slicers for Category, Sub-Category, Days, Rating Status, Segment, and State to filter and analyze the dashboard data.

Created three bookmarks Main View, Furniture – Tamil Nadu, and Accessories – Kerala to quickly switch between different dashboard views





9. Insights & Conclusions

Key Findings:

- Identified monthly sales trends.
- Found the best and lowest performing categories.
- Compared sales and Targets across different categories.
- Analyzed orders, returns, and shipping modes.

Descriptive Analysis:

Analyzed sales, target, quantity, and orders.

Diagnostic Analysis:

Identified the reasons for differences in sales and target.

Predictive Analysis:

Used past sales trends to understand future sales possibilities.

Prescriptive Analysis:

Suggested focusing on high-performing products and reducing returns.

10. Conclusions

The analysis helped to understand sales, profit, orders, and category performance. Power BI made it easy to visualize the data and identify useful business insights.